
ECON 312 — Intermediate Microeconomics

Handout: Utility Maximization

Fall 2026

Each case states a utility function, prices, and income. Solve with the method named in the case title.

Case 1. Cobb–Douglas (Lagrangian)

Utility $u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$. Prices $p_1 = 1$, $p_2 = 4$; income $m = 16$.

(a) Write the Lagrangian and state the three first-order conditions.

(b) Find the tangency condition, then solve for (x_1^*, x_2^*) .

(c) Report the utility attained and the value of λ .

Case 2. Quasilinear (Lagrangian)

Utility $u(x_1, x_2) = 2\sqrt{x_1} + x_2$. Prices $p_1 = 1$, $p_2 = 2$; income $m = 20$.

(a) Write the Lagrangian and state the three first-order conditions.

(b) Solve the first-order conditions for (x_1^*, x_2^*) .

(c) Report the utility attained.

Case 3. Cobb–Douglas (Method 1: $MRS = \text{Slope}$)

Utility $u(x_1, x_2) = x_1^{1/3} x_2^{2/3}$. Prices $p_1 = 1$, $p_2 = 2$; income $m = 36$.

(a) Compute MU_1 and MU_2 , and use them to write the MRS as a function of the bundle (x_1, x_2) .

(b) Impose $MRS = -p_1/p_2$ at the given prices and solve for x_2 in terms of x_1 .

(c) Substitute into the budget constraint, solve for (x_1^*, x_2^*) , and report the utility attained.